Factor Skyline Monograph

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The Factor Skyline: A Unified Architecture for Multiplicative Number Theory

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Front Matter

Preface
This monograph develops the Factor Skyline — a two-dimensional representation of the integers in which each integer n becomes a column of width $\mathrm{lpf}(n)$ and height $n/\mathrm{lpf}(n)$ — as a unified geometric framework for multiplicative number theory. The work synthesizes four streams of investigation: the architectural foundation (Part I), the correlation and randomness theories (Part II), the information-theoretic, dynamical, and universality theories (Part III), and the meta-structural analysis of the framework itself (Part IV).

The central claim is that a single function — the least prime factor — generates, through five emergent concepts (width, height, activation, coverage, escape), the full suite of classical results in prime distribution theory, together with new structural invariants that are invisible on the one-dimensional number line.

The Factor Skyline does not change the integers or contradict existing mathematics. It provides a new ontology — a new way of seeing — in which the mechanisms behind prime behavior become geometrically visible.

Overview of the Factor Skyline
Each integer $n \geq 2$ has width

$$w(n)=\mathrm{lpf}(n)$$

and height

$$h(n)=n / \mathrm{lpf}(n).$$

Columns are placed consecutively; the cumulative horizontal position is $x_{\mathrm{FS}}(n)$.
Primes contribute $\Delta x=1$ (narrow escape spires); composites contribute

$$\Delta x=\mathrm{lpf}(n)$$

(wide covered columns).

Prime squares p^2 are activation thresholds where new width-p layers enter the geometry.
Each layer covers a fraction $1/p$ of the remaining space.
What escapes all layers is prime.

The escape density is

$$D(p) = \prod_{q \le p} \left(1 - \frac{1}{q}\right)$$

and it governs the entire architecture.

Notation Guide

Below is the corrected notation table, with all symbols, Greek letters, superscripts, subscripts, and operators properly rendered.

Notation Table

Symbol	Meaning
$\mathrm{lpf}(n)$	Least prime factor of n
$w(n)$, $h(n)$	Width and height: $w(n) = \mathrm{lpf}(n)$, $h(n) = n/\mathrm{lpf}(n)$
$(x_{FS}(n), y_{FS}(n))$	FS-coordinates
$\Delta x(n)$	FS-x increment: 1 if n is prime; $\mathrm{lpf}(n)$ if composite
$D(p)$	Escape density: $D(p) = \prod_{q \le p} \left(1 - \frac{1}{q}\right)$
$p\#$	Primorial: $2 \cdot 3 \cdot 5 \cdots p$
$\varphi(p\#)$	Euler totient of $p\#$: $\varphi(p\#) = p\# \cdot D(p)$
$v_q(H)$	Number of distinct residues of a k-tuple H modulo q
$S_q(H)$	Survival factor: $S_q(H) = \frac{q - v_q(H)}{q}$
C_H	Hardy–Littlewood k-tuple constant
$H_b(x)$	Binary entropy: $H_b(x) = -x\log_2 x - (1-x)\log_2(1-x)$
$K(x)$	Kolmogorov complexity
$\omega(n)$	Number of distinct prime factors of n
$\Omega(n)$	Number of prime factors of n with multiplicity
$\tau(n)$, $\sigma(n)$	Divisor count and divisor sum
$\mu(n)$	Möbius function
$\theta(x)$, $\psi(x)$	Chebyshev functions
T_{FS}	First-order theory of the FS axioms
(A1)–(A4)	Universality axioms
(M1)–(M3)	Meta-axioms: order, multiplication, unique factorization

Architectural Map

Part I: ARCHITECTURE
Ch 1. Primitives & Coordinates
Ch 2. Templates & Epochs
Ch 3. Coverage Protection
Ch 4. Geometric PNT & Dickman
Ch 5. Möbius, Divisors, Conservation

Part II: STATISTICS
Ch 6. Correlation Theory
Ch 7. Randomness Theory
Ch 8. Unified Statistics

Part III: FOUNDATIONS
Ch 9. Entropy Theory
Ch 10. Ergodic Theory
Ch 11. Universality Theory

Part IV: META-THEORY
Ch 12. Meta-Structure
Ch 13. Meta-Mathematics
Ch 14. Unified Meta-Architecture

Part I: The Factor Skyline Architecture

Chapter 1. The Five Primitives

Definition I.1 (Least prime factor).
For $n \geq 2$,

$$\mathrm{lpf}(n)=\min\{\, p : p \mid n,\ p \text{ prime} \,\}.$$

Set lpf(1)=1.

Definition I.2 (Width and height).

$$w(n)=\mathrm{lpf}(n), \qquad h(n)=n / \mathrm{lpf}(n).$$

Definition I.3 (FS-coordinates).

$$x_{\mathrm{FS}}(1)=1, \qquad y_{\mathrm{FS}}(1)=1.$$

Define the horizontal increment:

$$\Delta x(n)=\{1,\ \ n \text{ prime},\ \ \backslash[4pt]\mathrm{lpf}(n),\ \ n \text{ composite}.$$

Then

$$x_{FS}(n)=x_{FS}(n-1)+\Delta x(n),$$

$$y_{FS}(n)=\{n,\ \text{n prime},\ \ \backslash[4pt](n / \mathrm{lpf}(n)),\ \text{n composite}.$$

Definition I.4 (Recursive width decomposition).
The column $W(n)$ is the sequence of widths obtained by iteratively applying lpf until reaching 1.
Its length is

$$\Omega(n)+1,$$

where $\Omega(n)$ is the total number of prime factors of n counted with multiplicity.

Definition I.5 (Activation).
A prime p activates at p^2.

Theorem I.1.

$$p^2=\min\{\, n:\mathrm{lpf}(n)=p \,\}.$$

Proof.
All multiples of p below p^2 have a smaller prime factor. ▮

Definition I.6 (Epoch).

$$\mathrm{Epoch}_k=[\, p_k^2,\ p_{k+1}^2 \,),$$

and coverage is frozen within each epoch.

Definition I.7 (Coverage and escape density).

$$D(p) = \prod_{\substack{q \le p \\ q \text{ prime}}} \left(1 - \frac{1}{q}\right)$$

Theorem I.2 (CRT independence).
Coverage layers are independent: for distinct primes $q_1,\ldots,q_k$, the probability that a random integer is coprime to all of them is

$$\prod_{i=1}^{k} \left(1 - \frac{1}{q_i}\right)$$

Proof. Chinese Remainder Theorem. ▮

Definition I.8 (Escape).
An integer n escapes if $\mathrm{lpf}(n)=n$; equivalently, n is prime.

Theorem I.3.
The escape set is exactly the set of primes.

Table I.1.
FS-coordinates for n=1 to 30 (see Appendix A).

Chapter 2. Templates and Epochs

Definition I.9 (Primorial and template).
The primorial of p is

$$p\#=2\cdot 3\cdot 5\cdots p.$$

The template T_p classifies each residue class modulo $p\#$ as open (coprime to $p\#$) or covered (divisible by some prime $q\leq p$).

Theorem I.4 (Template periodicity).
Width assignments modulo $p\#$ depend only on $n\ p\#$.

Theorem I.5 (Open-position count).

$$\varphi(p\#)=p\#\cdot D(p),$$

where

$$D(p)=\prod_{q\leq p}\left(1-\frac{1}{q}\right)$$

is the escape density.

Table I.2. Template statistics

$p\#$	Period	$\varphi(p\#)$	$D(p)$	Twin-open	Triplet-open
30	30	8	0.267	3	2
210	210	48	0.229	15	8
2310	2310	480	0.208	135	64
30030	30030	5760	0.192	1485	640

(All counts verified in Appendix A.)

Theorem I.6 (No activation within a square window).
The interval

$$[n2, (n+1)2)$$

contains no activation event in its interior.

Chapter 3. Coverage Protection and Template Persistence

Definition I.10 (Residue occupation number).
For a k-tuple $H=\{h_1,\ldots,h_k\}$,

$$v_q(H)=|\{ hi\, q \}|$$

is the number of distinct residues modulo q.

Definition I.11 (Admissibility).
A k-tuple H is admissible if

$$v_q(H)<q \qquad \text{for all primes } q.$$

Theorem I.7 (Survival factor).
For an admissible H,

$$S_q(H) = \frac{q - \nu_q(H)}{q}, \quad q > 0$$

Theorem I.8 (Coverage protection).
A width-q layer cannot eliminate all members of an admissible constellation.
If $q|(r+hi)$, then

$$q\nmid(r+hj) \qquad \text{for any } hj \not\equiv hi \pmod{q}.$$

Theorem I.9 (k-tuple constant).

For any admissible H,

$$C_H = \prod_q \frac{S_q(H)}{(1-1/q)^k} > 1$$

Table I.3. k-tuple constants

Constellation	C_H
Twin primes	$C_{\text{twin}} = 1.320$
Sexy primes	$C_{\text{sexy}} = 2.646$
Prime triplet	$C_{\text{triplet}} = 2.858$
Prime quadruplet	$C_{\text{quad}} = 4.151$

Theorem I.10 (Template persistence).
Let $NH(p)$ be the number of open placements of H in the template modulo $p\#$. Then

$$N_H(p_{k+1}) = N_H(p_k) \cdot (p_{k+1} - \nu_{p_{k+1}}(H))$$

Thus the constellation-open count per template period grows without bound.

Theorem I.11 (FS-x footprint invariance).
Every admissible constellation has a fixed FS-x inter-escape gap pattern, invariant across all sufficiently large instances.

Constellation	FS-x Gaps	Span	Verified Instances
Twin $(0,2)$	$[3]$	3	All 23 below 500
Triplet $(0,2,6)$	$[3,8]$	11	All 11 below 500
Quadruplet $(0,2,6,8)$	$[3,8,3]$	14	All 4 below 500

Chapter 4. The Geometric PNT and Dickman

Smoothness

Theorem I.12 (Activation horizon).
Every composite $n \le N$ satisfies

$$\mathrm{lpf}(n) \le \sqrt{N}.$$

Theorem I.13 (Geometric PNT).

$$\pi(N) \sim N \cdot D(\sqrt{N}).$$

Using Mertens' theorem,

$$\pi(N) = \Theta\,(N / \ln N).$$

Table I.4. PNT verification

N	$\pi(N)$	$N \cdot D(\sqrt{N})$	$N/\ln N$	$\pi(N)/(N/\ln N)$
1000	168	152.9	144.8	1.160
10000	1229	1203.2	1085.7	1.132
100000	9592	9651.9	8685.9	1.104

Theorem I.14 (Dickman function from column peeling).
Let $\Psi(x, x^{1/u})$ be the count of $x^{1/u}$-smooth numbers up to x. Then

$$\frac{\Psi(x, x^{1/u})}{x} \implies \rho(u)$$

where the Dickman function $\varrho(u)$ satisfies the delay-differential equation

$$u\,\rho'(u) = -\rho(u-1), \qquad u > 1.$$

Theorem I.15 (Smooth–rough duality).
The smooth density $\varrho(u)$ and the rough density

$$D(y) = \prod_{q \le y} \left(1 - \frac{1}{q}\right)$$

are dual views of the same underlying coverage product.

Chapter 5. Möbius, Divisors, and the Conservation Law

Theorem I.16 (Squarefree density).

$$\prod_p \left(1 - \frac{1}{p^2}\right) = \frac{6}{\pi^2}$$

Theorem I.17 (Möbius equidistribution).
Among squarefree integers, the Möbius function $\mu(n)$ is equidistributed between +1 and −1.

Theorem I.18.
Let $n=\prod p_i^{\wedge} e_i$. Then:

$$\tau(n) = \prod_i (e_i + 1)$$

counts sub-column selections, and

$$\sigma(n) = \prod_i \frac{p_i^{e_i+1} - 1}{p_i - 1}$$

sums sub-column heights.

Theorem I.19 (Erdős–Kac).

$$\frac{\omega(n) - \ln\ln n}{\sqrt{\ln\ln n}} \implies \mathcal{N}(0,1)$$

(From CRT independence + the Central Limit Theorem.)

Theorem I.20.

$$E[\tau(n)] \sim \ln N, \quad E\left[\frac{\sigma(n)}{n}\right] \longrightarrow \frac{\pi^2}{6}$$

Theorem I.21 (Conservation law).

$$\theta(x) = \sum_{p \le x} \ln p \sim x$$

Escape height accumulates at unit rate.

Table I.5. Conservation verification

x	$\theta(x)/x$
10^2	0.837
10^3	0.956
10^4	0.990
10^5	0.997

Theorem I.22 (Template width invariants).
The width-q column count per $p\#$-period is invariant:

$$n_q = \frac{p\#}{q} \prod_{r<q} \left(1 - \frac{1}{r}\right)$$

Part II: Correlations and Randomness in the Skyline

Part II builds on the architectural foundation of Part I to develop the statistical theory.
The key insight: all correlations and all apparent randomness arise from a single mechanism — CRT independence of width layers (Theorem I.2).

Chapter 6. Correlation Theory

6.1. The Master Decomposition

Definition II.1 (Layer status).
For a prime q and offset h,

status(q,h)={shared, q|h, \[4pt]independent, q∤h.

Theorem II.1 (Shared/independent decomposition).
For any multiplicative function f and offset h, the correlation $C_f(h)$ decomposes as follows:

- Shared layers ($q|h$) produce the correlation signal.
- Independent layers ($q\nmid h$) produce decorrelation.

6.2. Escape Correlations

Theorem II.2 (Hardy–Littlewood pair correlation).

$$\frac{C(h)}{C(2)} = \prod_{\substack{q|h \\ q\geq 3}} \frac{q-1}{q-2}$$

Corollary II.1 (Sexy-prime doubling).

$$C(6)=2\,C(2).$$

The factor 2 comes from $q=3$:

$$3-1\ /\ 3-2=2.$$

Table II.1. Pair counts to 10^5

h	Count	Ratio $C(h)/C(2)$	Predicted	Error
2	1224	1.000	1.000	—
6	2447	1.999	2.000	
30	3328	2.719	2.708	
210	3923	3.205	3.188	

All match predictions to within **2%.**

6.3. Parity Correlations

Theorem II.3 (Möbius correlation).
For the Möbius function $\mu(n)$:

- If $q \nmid h$: independent parity flips ⇒ zero cross-correlation.
- If $q \mid h$: correlated contribution of order $O(1/q^2)$.

Thus the total correlation satisfies:

$$R_\mu(h,N) \longrightarrow 0.$$

(This is the Chowla mechanism.)

Table II.2. Möbius correlations at $N=50000$
All values satisfy

$$|R_\mu(h)|<0.006.$$

4.1.4 6.4. Branching Correlations

Theorem II.4 (Divisor correlation).
Shared width layers inflate both $\tau(n)$ and $\tau(n+h)$, producing positive excess correlation.
Composite offsets have stronger correlations.

Theorem II.5 (ω non-mixing).

$$\operatorname{corr}(\omega(n), \omega(n+h)) \geq \frac{\sum_{q|h} \frac{1}{q}\left(1-\frac{1}{q}\right)}{\operatorname{Var}(\omega)} > 0$$

for all h with prime factors.
The correlation does not decay.

Table II.3. ω correlations at N=50000

h	Correlation
6	+0.371
30	+0.525
1000	+0.248

Correlations remain **persistently positive.**

4.1.5 6.5. Spectral Correlations

Theorem II.6 (Level repulsion).
Normalized zero spacings exhibit quadratic repulsion:

$$P(\text{spacing}<\varepsilon)=O(\varepsilon^2).$$

Incommensurate prime harmonics prevent mode degeneracy.

Table II.4. Normalized spacings (first 29 zeros)

- 0 spacings below 0.3
- 27 spacings in [0.3,1.5]
- 2 spacings above 1.5

This is GUE-consistent.

4.1.6 6.6. The Correlation Hierarchy

Theorem II.7 (Four-level hierarchy).

escape ⟶ parity ⟶ branching ⟶ spectral.

Each level subsumes the previous via Dirichlet series.

Theorem II.8 (CRT universality).

All four levels arise from CRT independence.

Chapter 7. Randomness Theory

4.2.1 7.1. The Sub-Poisson Escape Process

Theorem II.9 (Sub-Poisson).
The variance-to-mean ratio of prime counts in fixed-width windows satisfies

$$\text{VarMean} \approx 0.46,$$

compared to 1.0 for a Poisson process.
The deterministic template imposes spacers that suppress variance.

Table II.5. Sub-Poisson verification

Window W	Var/Mean
50	0.499
100	0.463
200	0.473

4.2.2 7.2. The Template/Stochastic Split

Theorem II.10 (73/27 split).
The FS-x increment sequence $\Delta x(n)$ is:

73.3% template-determined
(forced by n 30), and

26.7% stochastic
(open positions where $\Delta x=1$ or $\Delta x=\text{lpf}(n)$ with large least prime factor).

Table II.6. Δx autocorrelation

Lag	Autocorrelation
1	−0.100
6	+0.167
30	+0.212

All correlations are **entirely template-driven.**

4.2.3 7.3. The Randomness Hierarchy

Theorem II.11 (Three levels).
There are three intrinsic randomness levels:

- Level 0 (template): rigid, periodic, zero entropy.
- Level 1 (escape): pseudo-random, sub-Poisson, CRT-driven.
- Level 2 (spectral): GUE-random, maximal entropy given arithmetic constraints.

7.4. The Mixing Dichotomy

Theorem II.12 (ω/μ dichotomy).

- $\omega(n)$ is ergodic but not mixing
  (persistent shared-layer correlations).
- $\mu(n)$ is conjecturally mixing
  (independent sign flips destroy persistent correlations).

Corollary II.2.
The same CRT mechanism produces opposite mixing behavior.
The non-linearity of the observation function determines whether mixing occurs.

Chapter 8. Unified Statistical Architecture

8.1. The Entropy Budget

Theorem II.13 (FS entropy budget).
The entropy of the FS-x increment sequence $\Delta x(n)$ decomposes as:

$$H(\Delta x)=2.483 \text{ bits/int(total)}$$

$$I(\text{template})=1.700 \text{ bits/int(deterministic, 68.5\%)}$$

$$H(\text{escape + activation} \mid \text{template})=0.783 \text{ bits/int(stochastic, 31.5\%)}$$

The escape layer alone carries approximately

$$0.26 \text{ bits/int}$$

of irreducible uncertainty — the information-theoretic content of the parity barrier.

8.2. The Unified Randomness Principle
Theorem II.14 (FS randomness principle).
All apparent randomness in the skyline arises from CRT independence among coverage layers, amplified by multiplicative superposition.

<u>The primes are not random.</u>
They are deterministic escapes that appear random because CRT independence makes the escape process statistically indistinguishable from a filtered Bernoulli process.

Part II comprises 14 theorems, 1 definition, 2 corollaries, and 6 verification tables.
All results derive from CRT independence (Theorem I.2).

Part III: Information, Dynamics, and Universality

Part III examines the foundations of the statistical theory developed in Part II:
what information the skyline carries (entropy), how it evolves (dynamics), and which features are universal (universality).

Chapter 9. Entropy Theory

Theorem III.1 (Template: zero entropy, high information).

$$H(\text{template} \mid n\ p\#)=0.$$

The template removes 50–84% of primality uncertainty (Table III.1).

Theorem III.2 (Escape entropy).
The per-open-position entropy

$$Hb(D_{\text{open}})$$

peaks near $n\approx 10{,}000$, where $D\text{open}\approx 0.45$.

The per-integer escape entropy is

$$\approx 0.265 \text{ bits},$$

the irreducible core of primality uncertainty.

Theorem III.3 (Activation entropy).

$$H_{\text{activation}} = \sum_q H_b\left(\frac{1}{q}\right)$$

additive by CRT independence.

Among squarefree integers,

$$H(\mu \mid \text{squarefree})=1.000 \text{ bits},$$

the maximum possible binary entropy.

Theorem III.4 (Spectral entropy).
The GUE distribution achieves maximum entropy subject to the constraints imposed by prime harmonics.

Theorem III.5 (Parity barrier as information barrier).
Approximately

$$0.26 \text{ bits/int}$$

is information that the architecture cannot provide: the distinction between

$$\omega(n)=1 \quad \text{and} \quad \omega(n)\geq 3$$

at open positions.

This equals the conditional entropy:

$$H(\text{escape} \mid \text{template}).$$

Chapter 10. Ergodic Theory

Theorem III.6 (Template rigidity).
The template is periodic with period $p\#$.
It is ergodic (Cesàro averages converge) but not mixing (correlations do not decay).

Theorem III.7 (Escape ergodicity).

$$\pi(N) / N \to 0 (\text{ergodic}).$$

Under RH, the mixing rate is

$$O\,(1\sqrt{N}).$$

The escape process is super-ergodic relative to a Poisson process because variance is suppressed by the deterministic template.

Theorem III.8 (ω non-mixing, μ mixing).
Restating Theorems II.5 and II.12 in ergodic language:

- $\omega(n)$: **ergodic but not mixing**
  (persistent correlations from shared rigid components).

- μ(n): **conjecturally mixing**
  (parity flips destroy persistent correlations).

Theorem III.9 (Quasi-ergodicity).

The FS-x increment sequence Δx(n) is:

- 73% rigid,
- 27% mixing.

As n→∞:

- The system becomes asymptotically rigid because D(p)→0.
- But it remains permanently ergodic because D(p)>0 for all finite p.

Theorem III.10 (Entropy–ergodicity correspondence).

- Zero entropy ⇔ rigid.
- Positive entropy ⇔ ergodic.
- Maximum entropy ⇔ GUE-mixing.

The entropy content of a component determines its dynamical type.

Chapter 11. Universality Theory

Axioms (A1)–(A4)

- (A1) Countable generators with unique factorization (UF).
- (A2) Activation at g_k^2.
- (A3) CRT independence.
- (A4) Escape = complement of coverage.

Theorem III.11 (Three universality classes).

Class	Behavior of D	π-analogue	Example
I (Logarithmic)	$D(x) \sim C/\ln x$	$\pi(x) \sim x/\ln x$	Z
II (Constant)	$D(x) \to c > 0$	$\pi(x) \sim cx/n$	$F_q[x]$
III (Degenerate)	$D(x) \to 0$ fast	Finite escapes	$g_k = k$

Theorem III.12 (Twelve universal structures).
The following are consequences of axioms (A1)–(A4):

1. Product-form escape density.
2. Template periodicity.
3. PNT form: $\pi(x) \sim x\, D(x)$.
4. Erdős–Kac normality.
5. Möbius cancellation.
6. Squarefree density $=1/\zeta S(2)$.
7. Template persistence.
8. Coverage protection: $CH>1$.
9. Conservation law: $\theta(x) \sim x$.
10. Explicit-formula structure.
11. Sub-Poisson variance.
12. Epoch structure.

Theorem III.13 (GUE from incommensurability).
GUE statistics require:

- independent generators (Euler product),
- incommensurate logarithms ($\mathbb{Q}$-linear independence),
- minimal symmetry (functional equation).

Theorem III.14 (Universality criterion).
A phenomenon is universal iff it is provable from (A1)–(A4).
It is system-specific iff it requires generator values.

Theorem III.15 (Universal parity barrier).
The parity barrier is a consequence of (A1)–(A4) and applies to all systems in all universality classes.

5.3.1 11.6. The Renormalization Flow

Theorem III.16 ($D(p)$ as coupling constant).

$$D(p_{k+1}) = D(p_k)\left(1 - \frac{1}{p_{k+1}}\right)$$

The flow is:

- deterministic,
- irreversible,
- $D \to 0$ as $p \to \infty$.

The entire stochastic-to-deterministic transition is parameterized by D.

Theorem III.17 (Triple correspondence).
Entropy, ergodicity, and universality align:

Layer	Entropy	Ergodic Type	Universality
Template	0	Rigid	Universal
Escape	≈ 0.26 bits/int	Mixing	System-specific
Spectral	Max (GUE)	GUE-mixing	Universal (Class I)

Part III comprises 17 theorems.
The key result is the triple correspondence (Theorem III.17):

Information content determines dynamical behavior determines universality status.

Part IV: The Meta-Structure of the Factor Skyline

Part IV steps outside the framework to examine its own architecture:
its minimal assumptions, its structural symmetries, and its computational and logical properties.

Chapter 12. Meta-Structure

Theorem IV.1 (Primitive reduction).
All five FS concepts derive from the least prime factor alone:

$$\text{lpf} \longrightarrow \text{width} \longrightarrow \text{height,coverage} \longrightarrow \text{activation} \longrightarrow \text{escape}.$$

Theorem IV.2 (Three meta-axioms).
The function lpf
requires exactly three structural assumptions:

- (M1) Total order
- (M2) Multiplication
- (M3) Unique factorization

These axioms are minimal.

Theorem IV.3 (Emergence hierarchy).

Level	Structures
Primitive	lpf
First-order	Width, height, $D(p)$, templates
Second-order	PNT, Chebyshev functions, persistence, protection
Third-order	Explicit formula, zeros, GUE statistics

Theorem IV.4 (Three dualities).

Duality	Domain A	Domain B	Bridge
Coverage–Escape	Composites	Primes	Partition
Width–Spectral	Width layers	Zeta zeros	Explicit formula
Rigidity–Randomness	Template	GUE spectrum	Fourier analysis

Theorem IV.5 (Self-referential structure).
The framework's meta-structure mirrors its object-level structure:

- Width layers partition integers
  ↔ meta-layers partition results.
- Coverage limits escape
  ↔ the parity barrier limits provability.

The architecture describes not only the integers but the structure of the theory that describes the integers.

Chapter 13. Meta-Mathematics

Theorem IV.6 (Computability).
The skyline up to N is computable in

$$O(N\log\log N).$$

- Template statistics: O(k).
- Escape layer: **as hard as primality** (no faster method exists than determining primality itself).

Theorem IV.7 (Pseudo-randomness).
Let S_N denote the skyline up to N. Then:

$$K(SN)=O(\log N)$$

but

$$H(\text{escape} \mid \text{template}) \geq 0.26N.$$

Thus the escape layer is pseudo-random:

$$K \ll H.$$

A short program generates a sequence with large statistical entropy.

Theorem IV.8 (Barrier is methodological).
The parity barrier is not Gödel incompleteness.
It constrains coverage-based methods, not logic itself.

It is analogous to the impossibility of ruler-and-compass angle trisection:
a limitation of the method, not of mathematics.

Theorem IV.9 (T_{FS} as a first-order theory).
Axioms (A1)–(A4) define a first-order theory whose models include:

- Z
- Z[i]
- Fq[x]

The theory T_{FS} is incomplete because non-isomorphic models exist.

Theorem IV.10 (Universality = logical consequence).
A phenomenon is universal iff it is provable from T_{FS}.
This is the model-theoretic content of the universality criterion (Theorem III.14).

Theorem IV.11 (Computational–mathematical mirror).

Level	Math	Computation	Proof
Template	Coverage	$O(1)$/int	Trivial
Density	Averages	$O(\log N)$	Elementary
Escapes	Primes	$O(N \log\log N)$	Hard
Spectrum	Zeros	Unknown	Frontier

The computational hierarchy mirrors the mathematical hierarchy.

Chapter 14. Unified Meta-Architecture

Theorem IV.12 (Meta-correspondence).
Every meta-structural concept has a meta-mathematical counterpart:

Meta-Structure	Meta-Mathematics
lpf	Sieve program
Meta-axioms	T_{FS}
Emergence hierarchy	Computational hierarchy
Dualities	Compression / entropy
Renormalization	Increasing compression
Parity barrier	Incompressible core

These correspondences show that the architecture and its meta-theory are reflections of one another.

Theorem IV.13 (Boundary between architecture and logic).
The parity barrier admits four equivalent characterizations:

1. Information-theoretic:
   ≈0.26 bits/int that the template cannot provide.

2. Computational:
   The escape layer is as hard to compute as primality.

3. Logical:
   A gap between

$$\Sigma_1^0 \text{ (template persistence) and } \Pi_2^0 \text{ (escape persistence).}$$

4. Universal:
   The same boundary appears in all systems satisfying (A1)–(A4).

These four descriptions identify one and the same structural limit.

Part IV comprises 13 theorems.
The key insight: the framework's limits are as revealing as its successes —
the parity barrier tells us exactly what a proof of the open conjectures must supply.

Back Matter

Glossary

Term	Definition	First Appearance
Activation	Entry of width-p coverage at p^2	Def. I.5
Admissible	$v_q(H) < q$ for all primes q	Def. I.11
Column	Recursive width decomposition of n	Def. I.4
Coverage	Cumulative width-layer claiming	Def. I.7
Coverage protection	Width-q cannot hit all members of an admissible tuple	Thm. I.8
Epoch	Interval $[p_k^2, p_{k+1}^2)$; frozen coverage	Def. I.6
Escape	n with $\mathrm{lpf}(n) = n$ (prime)	Def. I.8
Escape density	$D(p) = \prod(1 - 1/q)$	Def. I.7
FS-x gap invariant	Fixed FS-x inter-escape pattern for each constellation	Thm. I.11
k-tuple constant	$C_H = \prod S_q(H)/(1 - 1/q)^k$	Def. I.10, Thm. I.9
Parity barrier	$\approx$ 0.26 bits/int that coverage cannot supply	Thm. III.5, IV.13
Primorial	$p\# = 2 \cdot 3 \cdots p$	Def. I.9
Quasi-ergodic	73% rigid + 27% mixing	Thm. III.9
Renormalization	Template extension $p_k\# \to p_{k+1}\#$; D as coupling	Thm. III.16
Sub-Poisson	Var/Mean < 1 for escape counts	Thm. II.9
Survival factor	$S_q(H) = (q - v_q(H))/q$	Thm. I.7
Template	Periodic open/covered classification mod $p\#$	Def. I.9
Template persistence	Constellation-open count grows without bound	Thm. I.10

Theorem Index

Part I — Architecture

Theorems: I.1–I.22

Definitions: I.1–I.11

Part II — Statistics

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Part IV — Meta-Theory
Theorems: IV.1–IV.13

Totals Across the Monograph
66 theorems
12 definitions
2 corollaries
4 major parts

This index provides a complete structural map of the logical architecture.

Comprehensive Reference List

[1] A. Proxmire, The Factor Skyline: An Ontological Lookout Over the Integers (2026). DOI: 10.5281/zenodo.18275273.

[2] H. Davenport, Multiplicative Number Theory, 3rd ed., Springer, 2000.

[3] G. H. Hardy and E. M. Wright, An Introduction to the Theory of Numbers, 6th ed., Oxford University Press, 2008.

[4] H. Iwaniec and E. Kowalski, Analytic Number Theory, AMS Colloquium Publications, 2004.

[5] F. Mertens, "Ein Beitrag zur analytischen Zahlentheorie," J. Reine Angew. Math. 78 (1874), 46–62.

[6] N. G. de Bruijn, "On the number of positive integers ≤ x and free of prime factors > y," Indag. Math. 13 (1951), 50–60.

[7] G. H. Hardy and J. E. Littlewood, "Some problems of 'Partitio numerorum'; III," Acta Math. 44 (1923), 1–70.

[8] P. Erdős and M. Kac, "The Gaussian law of errors in the theory of additive number theoretic functions," Amer. J. Math. 62 (1940), 738–742.

[9] H. Cramér, "On the order of magnitude of the difference between consecutive prime numbers," Acta Arith. 2 (1936), 23–46.

[10] A. Granville, "Harald Cramér and the distribution of prime numbers," Scand. Actuarial J. (1995), 12–28.

[11] H. L. Montgomery, "The pair correlation of zeros of the zeta function," Proc. Sympos. Pure Math. 24 (1973), 181–193.

[12] N. M. Katz and P. Sarnak, Random Matrices, Frobenius Eigenvalues, and Monodromy, AMS, 1999.

[13] P. Sarnak, “Three lectures on the Möbius function, randomness and dynamics,” 2011.

[14] S. Chowla, The Riemann Hypothesis and Hilbert’s Tenth Problem, Gordon and Breach, 1965.

[15] K. Gödel, “Über formal unentscheidbare Sätze der Principia Mathematica,” Monatsh. Math. Phys. 38 (1931), 173–198.

[16] A. N. Kolmogorov, “Three approaches to the quantitative definition of information,” Problems Inform. Transmission 1 (1965), 1–7.

[17] A. Weil, “Sur les courbes algébriques et les variétés qui s’en déduisent,” Actualités Sci. Ind. 1041 (1948).

7.4 Appendix A: Numerical Tables and Reproducibility

All numerical computations were performed using SymPy (Python) with the coordinate system implemented in FS_coordinates.py.
Every table in the text is fully reproducible using the scripts in the project repository:

https://github.com/allen-proxmire/factor-skyline

The appendix contains the complete numerical record supporting the architectural, statistical, and meta-structural results of the monograph.

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